

Retail Business Performance & Profitability Analysis Report

Prepared by: Suman

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1. Abstract

This project presents an end-to-end data analysis pipeline designed to identify profit-draining components within a multi-variable retail operations environment. By preparing an active transactional dataset, identifying operational metrics, and mapping visual cross-dependencies, this study uncovers a significant structural liability: a strong negative correlation (-0.6842) between product shelf-time and transaction profitability. The findings isolate specific negative margin categories (Footwear at -6%) and establish clear, data-driven parameters for inventory optimization and procurement adjustments.

2. Introduction

Modern retail management relies heavily on minimizing operational inefficiencies within supply chains. Excessive inventory shelf-time ties up capital and elevates storage costs, directly dampening profit performance. This project aims to track and evaluate these challenges by simulating structural transactional metrics, cleaning missing variables, performing statistical evaluations, and constructing a responsive business intelligence dashboard.

3. Tools Used

- **Python:** Selected for data synthesis and foundational statistical discovery.
- **Power BI (Power Query & DAX):** Utilized as the primary extract, transform, load (ETL) platform, metric calculation engine, and user-facing reporting layer.
- **Microsoft Edge:** Employed to inspect, verify, and export the static presentation formats.

4. Steps Involved in Building the Project

- **Step 1: Data Synthesis:** Constructed a multi-variable transactional profile containing dimensions for region, season, product sub-category, financial costs, and inventory velocity.
- **Step 2: Data Cleansing (ETL):** Loaded the dataset into Power BI Power Query. Filtered fields to remove blank and null parameters to establish absolute data integrity.
- **Step 3: Statistical Evaluation:** Conducted data checks to establish the exact relationship between velocity metrics and net gains, confirming a downward regression curve.
- **Step 4: Dashboard Engineering:** Developed the UI using Power BI Desktop, mapping global slicers, dynamic KPI totals, and multi-axis charts with integrated conditional formatting rules.

5. Conclusion & Strategic Suggestions

The data analysis confirms a severe operational bottleneck: a strong negative correlation (-0.68) proving that the longer inventory sits on shelves, the more it

destroys net profitability due to rising holding costs. To aggressively reduce Average Inventory Days and recover tied-up capital, the business must deploy the following three-part strategy:

- **Immediate Inventory Liquidation:** The **Footwear** sub-category is currently a major profit drain showing a net margin of **-6%**. The company should immediately implement a dynamic markdown cadence (e.g., 20% discount after 60 days on the shelf, 40% after 90 days) and use loss-leader bundling (pairing slow-moving Footwear with high-performing Clothing tops) to clear warehouse space rapidly.
 - **Merchandising & Marketing Enhancements:** Lagging products must be given a visibility boost. This includes moving overstocked items to high-traffic physical store zones (front windows and end-caps) and launching targeted digital remarketing campaigns specifically aimed at the geographic regions (North, South, East, West) where the dashboard shows sales are slowest.
 - **Supply Chain & Demand-Driven Purchasing:** To prevent future overstocking cycles, the procurement team must transition away from bulk-ordering. Moving to a Just-In-Time (JIT) inventory model and setting automated Reorder Points (ROP) in Power BI based on active regional sales velocity will ensure inventory matches actual customer demand.
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